Run report: E8 and E9a, per-system F4 profile

2026-10-18, 17:42–18:15 UTC. research/isogeny_conductor_gap_round6_20261018.

6,000 S_3 descent systems: 5 Koblitz curves (a = 1 over F_2^{17}, F_2^{19}, F_2^{23} a = 0 over F_2^{19}, F_2^{23}), default and trace-coset bases at d = 8, 9, 10, with 200 shared targets. Each system was solved by full f4-f2 and by f4-f2-early (free = 8). Work counts are deterministic.

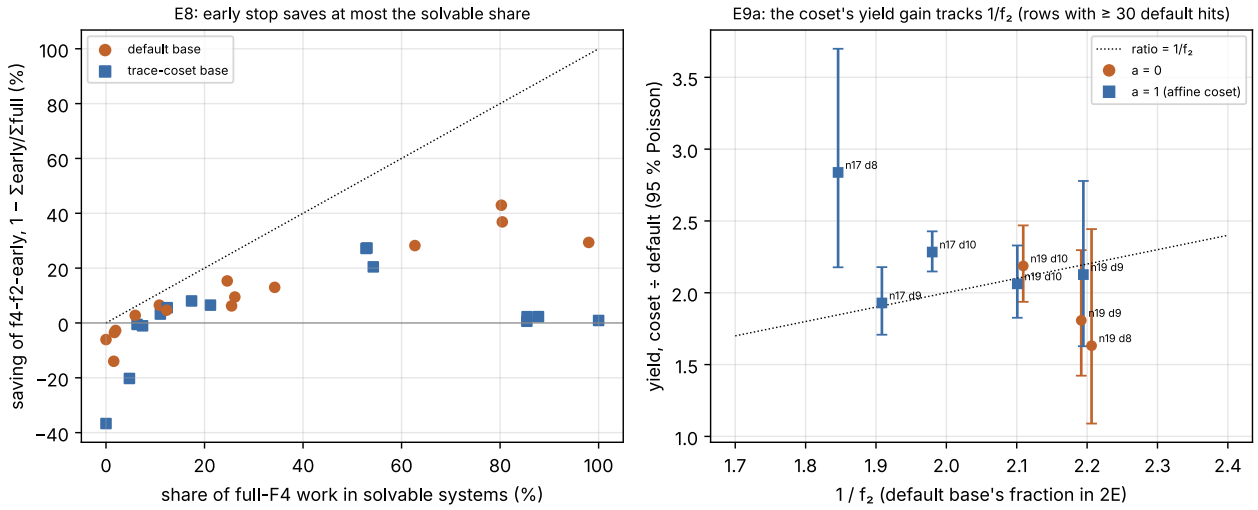

Figure 1: Left: early-stop saving against the solvable share of work (P-E8d). Right: coset yield gain against $1/f_2$ (P-E9a).

- P-E8b, exactness: holds. Zero mismatches.
- P-E8a, solvable at least 3 times unsolvable: fails. Mostly 1.0 to 2.3.
- P-E8c, early stop at most 0.5 of full on solvable systems: fails pooled, at 0.94 [0.93, 0.95].
- P-E8d, saving at most the solvable share: holds. Savings are up to 43 % and down to −37 %.
- P-E9a, yield ratio near $1/f_2$: holds in 7 of 9 rows. The F_2^{19} explanation is refuted: f_2 there is 0.45.
- P-E9b: holds in 7 of 9 rows.
- P-E9c: holds in 8 of 9 rows. The coset is 34 to 45 times costlier where the solving degree rises to 4, including the linear a = 0 case.